

G+1+TERRACE RCC RESIDENTIAL BUILDING

TWO-UNIT DUPLEX – SCHEMATIC ARCHITECTURAL & STRUCTURAL DRAWING SET

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A-000	Cover / Index / Design Basis Notes	S-004A	Typical Reinforcement Details (Rebar Sizes)
A-001	Site Development Plan	S-004B	Steel Quantity Estimate (Preliminary BBS Summary)
S-001	Column Layout Plan	S-004C	Preliminary Material & Cost Estimate
S-002	Foundation Layout Plan	A-005	Staircase Plan & Section
A-002	Ground Floor Plan	P-001	Plumbing Layout (Water Supply, Soil & Waste)
A-003	First Floor Plan	A-006	Building Sections A-A & B-B
A-004	Terrace Floor Plan	A-007	Front Elevation (North)
S-003	Beam Layout Plan (GF Roof / FF Slab)	A-008	Side Elevation (East)
S-004	Slab Layout & Reinforcement Direction Plan	A-009	Dimensioned Ground-Floor Blueprint (Master Sheet)

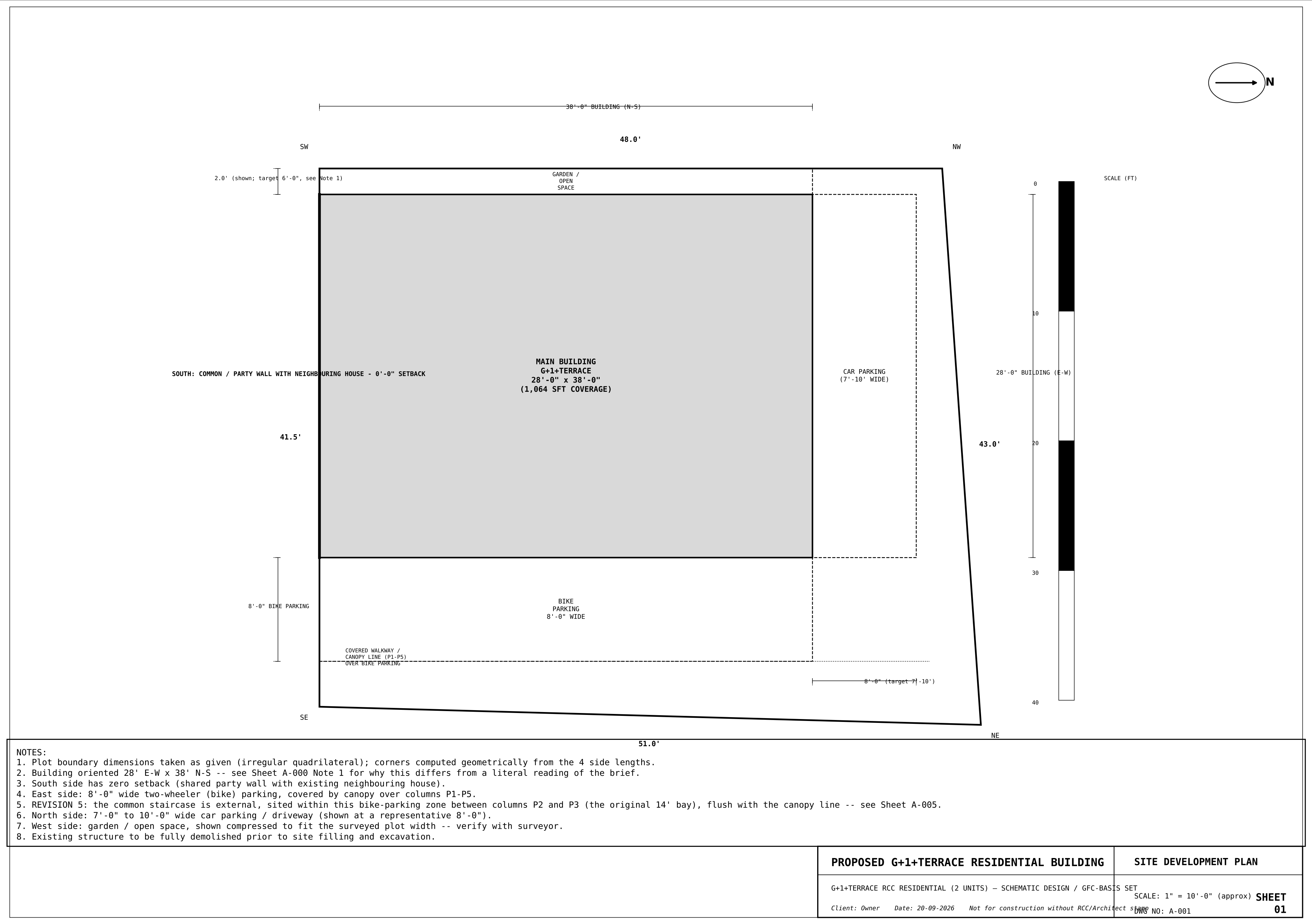
PROJECT DATA

- Plot (as surveyed): East 51.0' | West 48.0' | North 43.0' | South 41.5' (irregular quadrilateral)
- Building footprint: 28' (E-W) x 38' (N-S) = 1064 sq.ft. ground coverage
- Structure: RCC framed, isolated footings, 18 columns 12"x15", RCC beams & slab, brick/block infill walls
- Floors: Ground Floor (2 residential units) + First Floor (4 bedrooms + common facilities) + Terrace
- Soil: assumed rocky, existing structure demolished, 3'-0" compacted filling above existing ground level
- Electrical layout: EXCLUDED from this set -- to be provided separately by the client.

DESIGN BASIS / RECONCILIATION NOTES (read before use):

- The brief's site boundary lengths (E 51', W 48', N 43', S 41.5') and the requested building (38'x28') + setbacks (E 8' bike, W 6' garden) do not reconcile if the building's 38' side is read East-West: 38+8+6=52" would exceed the ~42' plot width on that axis. Reconciliation: the building is oriented with its 28' side East-West and 38' side North-South (28+8+6=42', matching the plot width almost exactly). Total main footprint unchanged at 28x38=1,064 sq.ft. Re-verify against the actual survey before construction.
- REVISION 2: Ground floor rebuilt to match the client's own layout PDF -- two 12'-deep end units (Bedroom + Living Room, South and North) flank a full-depth 14' wide central corridor (the 'P2-P3' grid bay) that houses the common stair plus each unit's own WC, Shower and Kitchen.
- 18 columns P1-P18. P1-P5 (canopy line, spacing 12-14-12-9) sit 8' east of the main east face, over the bike-parking walkway. P6-P9/P11-P14/P15-P18 are the three main N-S lines (spacing 12-14-12). P10 -- not part of any stated bay -- is the corner column between P9 and P5 that carries the first-floor gallery's extension over the car-parking side.
- REVISION 4/5: Common staircase relocated OUTSIDE the main building, per client feedback -- an open/covered external dog-leg stair in the bike-parking zone (x=28-36, between the main wall and the canopy line), positioned between columns P2 and P3 -- the original 'P2-P3' bay, y=12-26, 14' available, exactly as marked up by the client. 6' wide, two 3' flights side by side, 18 risers @ 7" / 16 treads @ 10.5" for the 10'-6" floor height, 5' landing. The former internal stair void is removed from the central corridor, which is now a plain spine (Kitchen widened to fill the freed space) -- see Sheets A-002, A-003, A-004 and A-005.
- REVISION 2: First floor rebuilt to match the client's sketch -- 4 rooms (14'x9'-6" each, per the original brief) along the West/garden side, each with a 6'-deep ensuite bath and opening onto a CONTINUOUS 6' deep back balcony spanning the full West side. A gallery/lounge along the East side cantilevers out to fully cover the 8' bike-parking width below, and extends north along it to land squarely on column P10 (directly above P9, in line with P5), covering the North car-parking setback in the same reach.
- REVISION 3: The P11-P14 column line moved from x=12' to x=18' (and first-floor room depth reduced 14'-->12') after the client noticed a column landing inside the first-floor rooms. x=18' is exactly the Room/Gallery wall on the first floor AND the Kitchen's west wall on the ground floor -- as a bonus this also fixes a pre-existing problem where x=12' sat inside the stair void itself. See Sheets S-001 and A-003.
- REVISION 6: Electrical layout REMOVED from this set at the client's request (to be supplied separately). Added Sheets S-004A (typical reinforcement details -- bar sizes/spacing for every footing, column, beam, slab and stair) and S-004B (a preliminary steel quantity estimate). Both are THUMB-RULE / preliminary sizing for budgeting and coordination only -- not a substitute for a full structural analysis and bar bending schedule (BBS), which remains required before construction.
- REVISION 7: 9 of the 18 footings (P1, P2, P3, P4, P5, P6, P10, P11, P15) sit at or very near a plot boundary and are HALVED on the boundary side(s), per client instruction, trimmed to either a 9 sq.ft quarter or an 18 sq.ft half footing (see REVISION 8 below for the final split between the two). See G.FOOT.TRIM / G.footing extents() in this file (the single source of truth used by Sheets S-002 and S-004A), and Sheet S-004A for the reinforcement detail. Plinth/tie beams at these 9 columns also act as STRAP BEAMS resisting the eccentric loading -- to be confirmed by the structural engineer. Sheet S-004B's steel quantity estimate reflects the reduced footing sizes.
- REVISION 8: Per further client instruction: (1) bath resized to 5'x6', total first-floor room (bath+room) resized to 12'x9'-6", which moves the P11-P14 column line to x=12' (was 18') -- giving P11-P15/P12-P16/P13-P17/P14-P18 = 12' and P6-P11/P7-P12/P8-P13/P9-P14 = 16', both as requested. (2) P5 reclassified as a true corner footing (quarter-size, kept inside both the east and north boundary lines), and P15 reclassified as a half footing like P11 (its west side has a full 6' garden setback, so only the south/party-wall side needs trimming) -- see G.FOOT.TRIM. (3) External staircase width reduced 6'-->3'; the dog-leg (two parallel 3' flights) is replaced by a single 3'-wide half-turn flight sharing one lane, see Sheet A-005. (4) Terrace slab's north extension corrected to cover only the gallery CANTILEVER width (x=28-36, landing on P10), not the full width back to the P11-P14 line -- removes the extra slab area that had been shown between P10 and P14 on both the First Floor and Terrace plans.
- REVISION 9: Ground Floor doors reviewed against the client's own layout reference (LAYOUT 26). Added a 3'-0" back door between each Bedroom and its WC (washroom), giving direct ensuite-style access without passing through the Living Room; swings into the Bedroom, clear of the WC fixture. Confirmed each unit's MAIN ENTRANCE remains on the Living Room (hall) East wall, per client instruction. See Sheet A-002.
- REVISION 10: Added Sheet S-004C, a preliminary MATERIAL & COST ESTIMATE (steel, cement, sand, aggregate/khadi, bricks, binding wire) covering the whole building, derived from this same geometry and from Sheet S-004B's steel BBS. Rates used are INDICATIVE ONLY (approx. national-average India) and exclude labour, shuttering, doors/windows, finishes and site overheads -- see S-004C for the full basis and exclusions.
- REVISION 11: Added LABOUR to Sheet S-004C, using client-supplied rates: RCC labour Rs 220/sqft (x total RCC/slab area cast) and general labour Rs 1100/labour-day. The number of general-labour-days is this set's own estimate (CPWD-type productivity: ~1.0 cum/day brickwork, ~16 sqm/day plastering, applied to the brickwork/plaster quantities already on S-004C) -- NOT a client-supplied figure, verify with the actual contractor. New grand total = material + labour, still excluding shuttering, finishes, other trades and site overheads.
- REVISION 12: Added a third labour line to Sheet S-004C: a material transformation/slab-filling crew (4 labour @ Rs1100/day + 4 labour @ Rs400/day = Rs6000/day), per client instruction. Day count estimated from the 96 cum RCC concrete volume at a thumb-rule 4 cum/day placement rate for this crew -- NOT a client-supplied figure. NOTE: this may overlap with the existing RCC labour line (Rs220/sqft), which already covers concreting; confirm with the contractor whether both are genuinely separate cost pools before totalling the budget.
- This package is a coordinated SCHEMATIC / design-development set for planning and client review. Final footing sizes, reinforcement, bar bending schedules and the cantilever design at the gallery must be verified and stamped by a licensed structural engineer before construction, notwithstanding the 'rocky soil' assumption used here.

PROPOSED G+1+TERRACE RESIDENTIAL BUILDING	COVER / INDEX / DESIGN BASIS NOTES
G+1+TERRACE RCC RESIDENTIAL (2 UNITS) – SCHEMATIC DESIGN / GFC-BASIS SET Client: Owner Date: 20-09-2026 Not for construction without RCC/Architect stamp	SCALE: NTS SHEET 00 DWG NO: A-000



- NOTES:
- Plot boundary dimensions taken as given (irregular quadrilateral); corners computed geometrically from the 4 side lengths.
 - Building oriented 28' E-W x 38' N-S -- see Sheet A-000 Note 1 for why this differs from a literal reading of the brief.
 - South side has zero setback (shared party wall with existing neighbouring house).
 - East side: 8'-0" wide two-wheeler (bike) parking, covered by canopy over columns P1-P5.
 - REVISION 5: the common staircase is external, sited within this bike-parking zone between columns P2 and P3 (the original 14' bay), flush with the canopy line -- see Sheet A-005.
 - North side: 7'-0" to 10'-0" wide car parking / driveway (shown at a representative 8'-0").
 - West side: garden / open space, shown compressed to fit the surveyed plot width -- verify with surveyor.
 - Existing structure to be fully demolished prior to site filling and excavation.

PROPOSED G+1+TERRACE RESIDENTIAL BUILDING

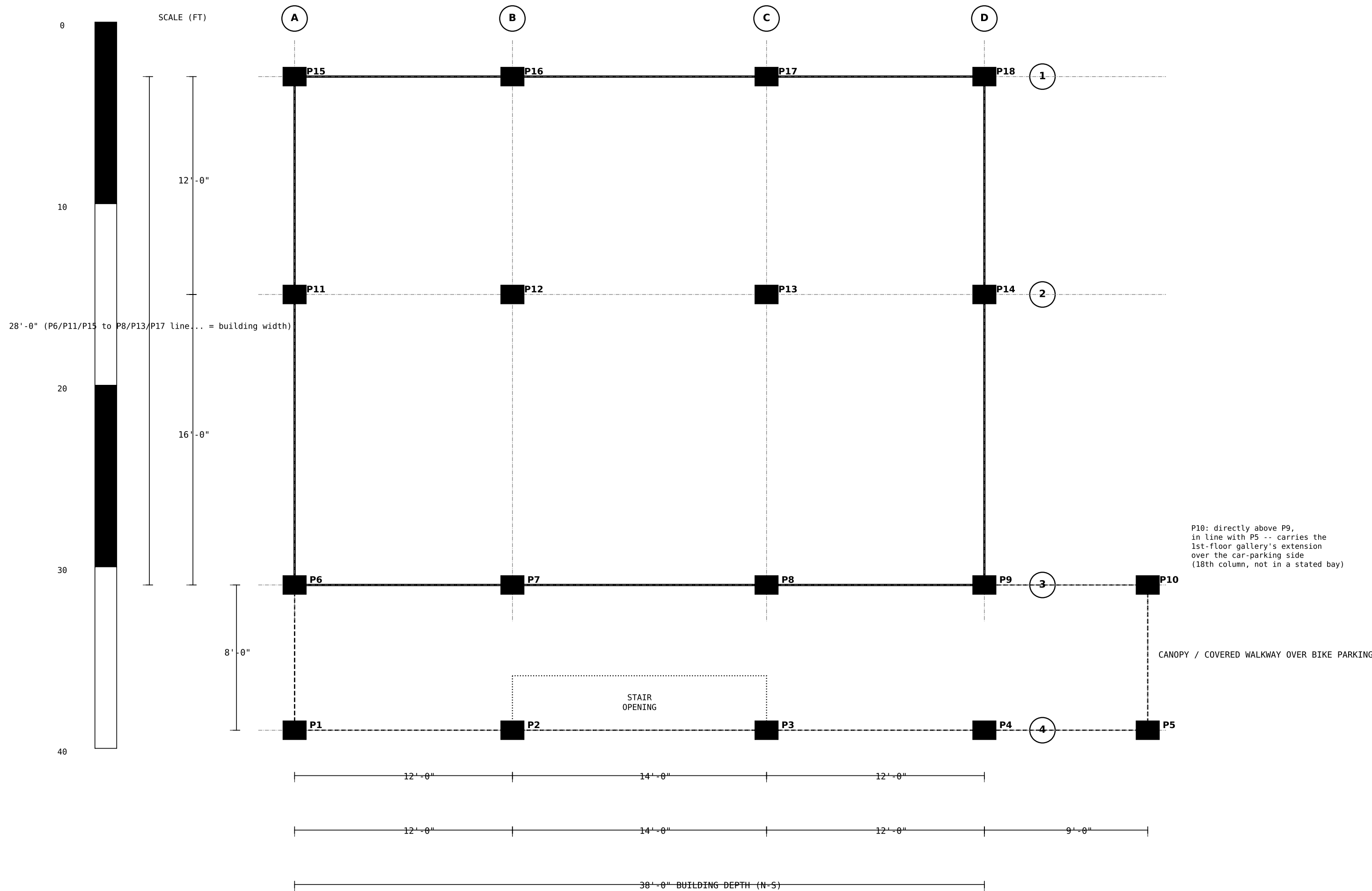
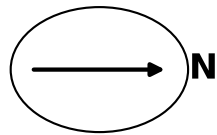
SITE DEVELOPMENT PLAN

G+1+TERRACE RCC RESIDENTIAL (2 UNITS) – SCHEMATIC DESIGN / GFC-BASIS SET

SCALE: 1" = 10'-0" (approx) SHEET

Client: Owner Date: 20-09-2026 Not for construction without RCC/Architect stamp DWG NO: A-001

01



- NOTES:
- 18 RCC columns, P1-P18, size 12" x 15" (0.3m x 0.38m) throughout, unless noted.
 - Grid line 4 (P1-P5) is the canopy line over the east bike-parking/entrance walkway, 8'-0" east of grid line 3.
 - Grid spacing: 12'-0" / 16'-0" (E-W, all rows); 12'-0" / 14'-0" / 12'-0" (N-S, rows A-D); canopy row adds a 9'-0" bay (P4-P5).
 - P10 is a supplementary corner column between P9 and P5, carrying the first-floor gallery's cantilevered extension over the car-parking side; see Sheet A-000.
 - All columns to be centred on grid lines shown; verify against structural design/BBS before construction.

PROPOSED G+1+TERRACE RESIDENTIAL BUILDING

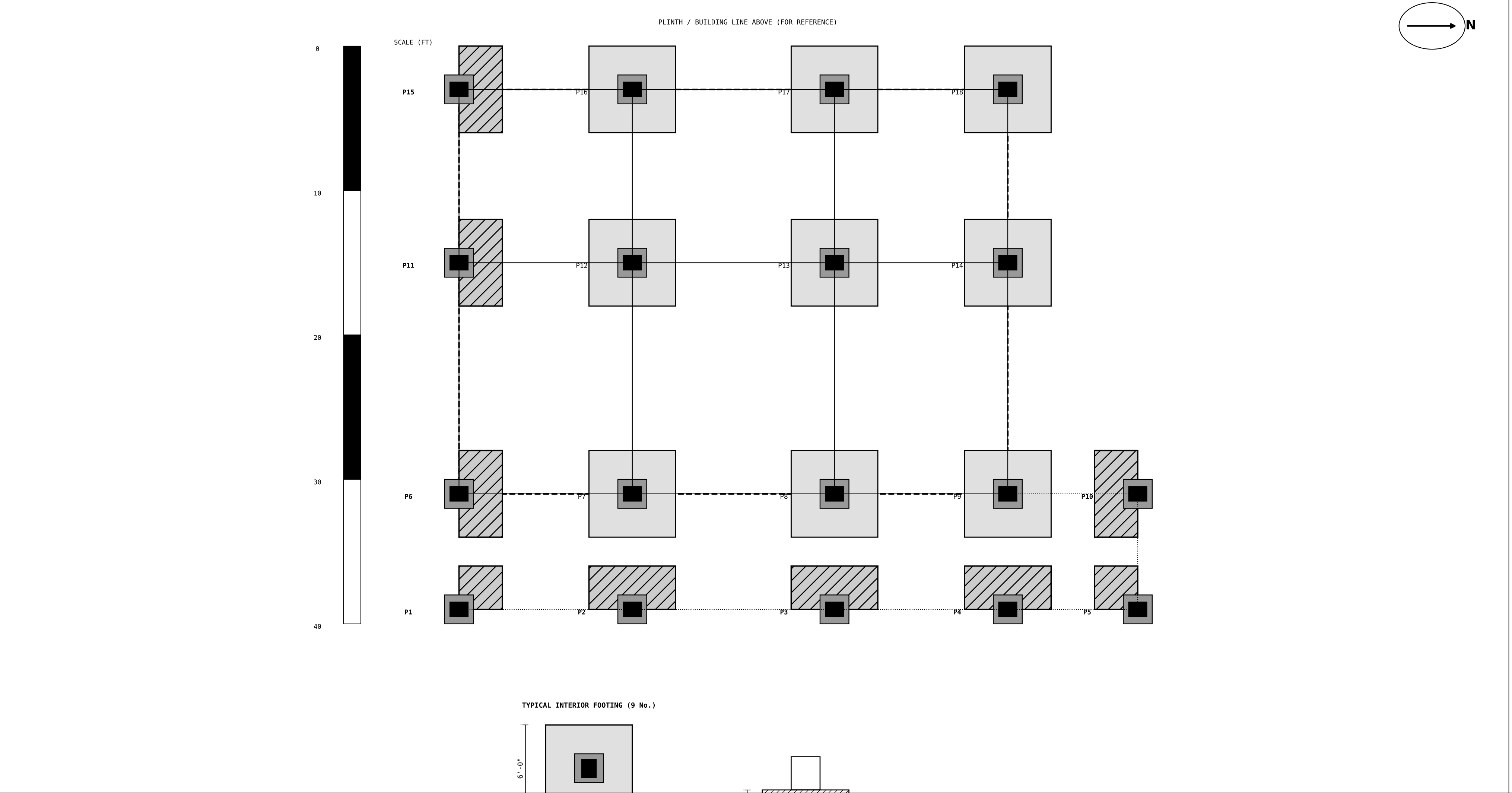
COLUMN LAYOUT PLAN (P1-P18)

G+1+TERRACE RCC RESIDENTIAL (2 UNITS) – SCHEMATIC DESIGN / GFC-BASIS SET

SCALE: 1:100

Client: Owner Date: 20-09-2026 Not for construction without RCC/Architect stamp DWG NO: S-001

SHEET
02



NOTES:

1. Soil condition: rocky (assumed) -- founding level to be confirmed on site by geotechnical inspection.

2. Existing house fully demolished; site filled with 3'-0" compacted (mechanically consolidated) fill above existing ground level before excavation.

3. REVISION 7: 9 of the 18 footings sit at or very near a plot boundary (south party wall, east canopy line, or north corner near P10/P5) and are HALVED on the boundary side(s), per client instruction -- hatched in plan, detailed on Sheet S-004A.

4. REVISION 8: P1 and P5 are true corners (halved BOTH ways, 9 sq.ft); P2, P3, P4, P6, P10, P11, P15 are halved on one side only (18 sq.ft) -- P15's west side has a full 6' garden setback so only its south side needs trimming.

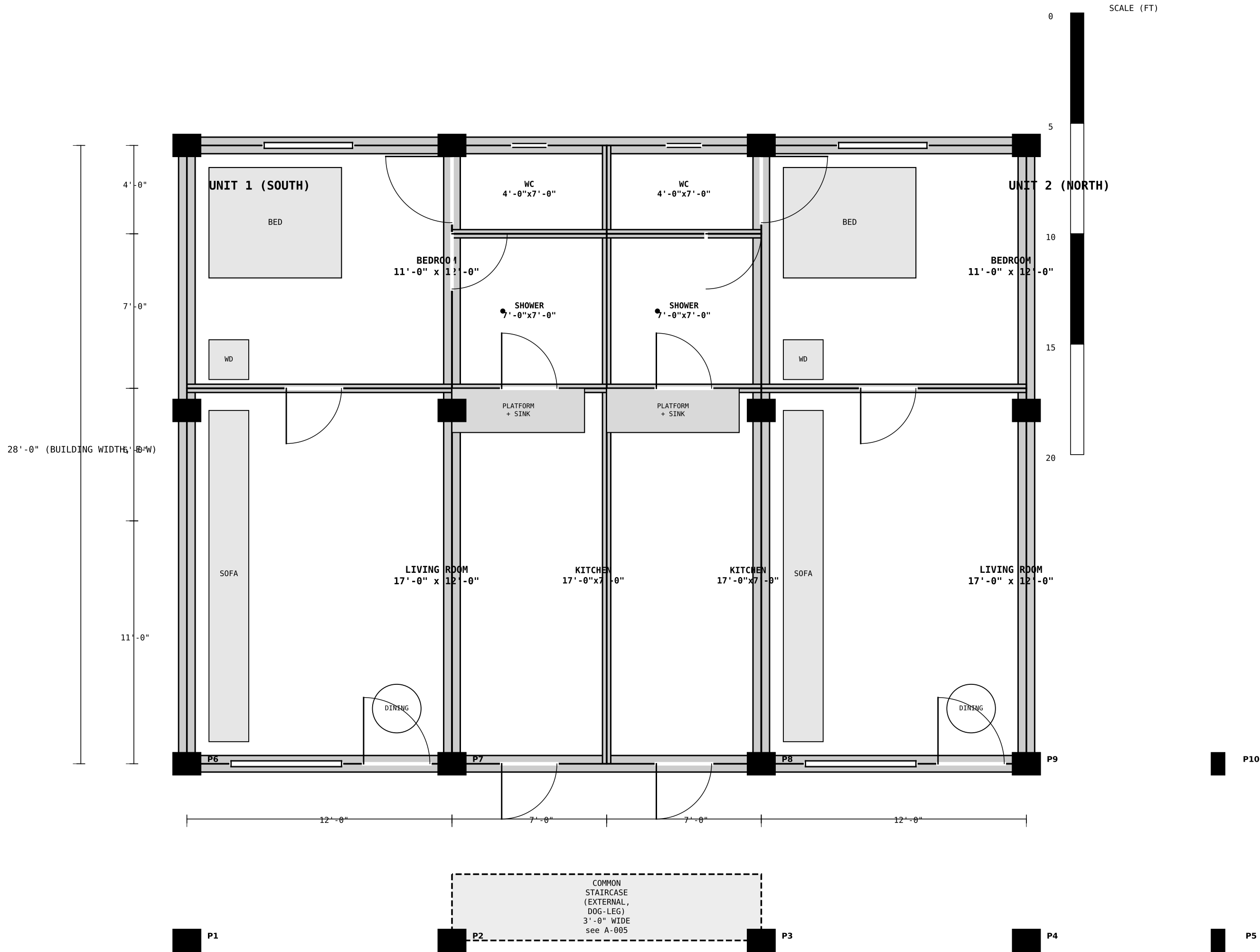
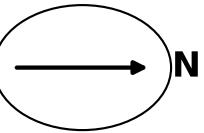
5. The remaining 9 footings (P7, P8, P9, P12, P13, P14, P16, P17, P18) are standard 6x6 sq. ft. HALF (P2,P3,P4, P6, P10, P11, P15) and CORNER (P1-P5)

6. Plinth/tie beams schematically link all footings along grid lines; for the 9 boundary footings these also act as STRAP BEAMS resisting the eccentric moment -- confirm sizing with the structural engineer.

7. P.C.C. (1:4:8) levelling bed, minimum 75mm (3"), below every footing. Pedestal 2'x2' RCC up to plinth level, same for all 18.

8. Footing sizes shown are schematic/typical; final sizing (including the reduced boundary footings) must be checked against actual safe bearing capacity (SBC) from a soil test report, the total column loads, and the eccentricity/strap-beam design, by the s

PROPOSED G+1+TERRACE RESIDENTIAL BUILDING		FOUNDATION LAYOUT PLAN (ISOLATED)	
G+1+TERRACE RCC RESIDENTIAL (2 UNITS) – SCHEMATIC DESIGN / GFC-BASIS SET		SCALE: 1:100	SHEET
Client: Owner Date: 20-09-2026 Not for construction without RCC/Architect stamp		DWG NO: S-002	03



- NOTES:
- REVISION 5: Common staircase moved OUTSIDE the building -- an external stair in the bike-parking zone between P2 and P3, the original 14' bay (see A-005). Central corridor is now a plain, unbroken spine.
 - Layout matches the client's own ground-floor sketch: two 12'-deep end units (Bedroom+Living, South and North) flanking a full-depth 14' central corridor.
 - The 14' central corridor carries each unit's own WC (4'x7'), Shower (7'x7') and Kitchen, now widened to 17'x7' after removing the internal stair void.
 - External walls 9" thick; internal partitions 4.5" thick; corridor/unit dividing walls 9" (load-bearing, carry the floors above).
 - Each unit's MAIN ENTRANCE is on the Living Room (hall) East wall, plus a second door from its Kitchen's East wall giving direct access to the external common staircase.
 - REVISION 9: Added a 3'-0" back door/passage between each Bedroom and its WC (washroom), for direct ensuite-style access without passing through the Living Room; swings into the Bedroom, clear of the WC fixture.
 - Door sizes: unit entry 3'-0"x7'-0", back door (Bedroom-WC) 3'-0"x7'-0", other internal 2'-6"x7'-0". Window sizes: bedroom 4'-0"x4'-0", living room 5'-0"x4'-6". Ventilators 1'-6"x1'-6" to WC/Shower.

PROPOSED G+1+TERRACE RESIDENTIAL BUILDING

GROUND FLOOR PLAN (TWO RESIDENTIAL UNITS)

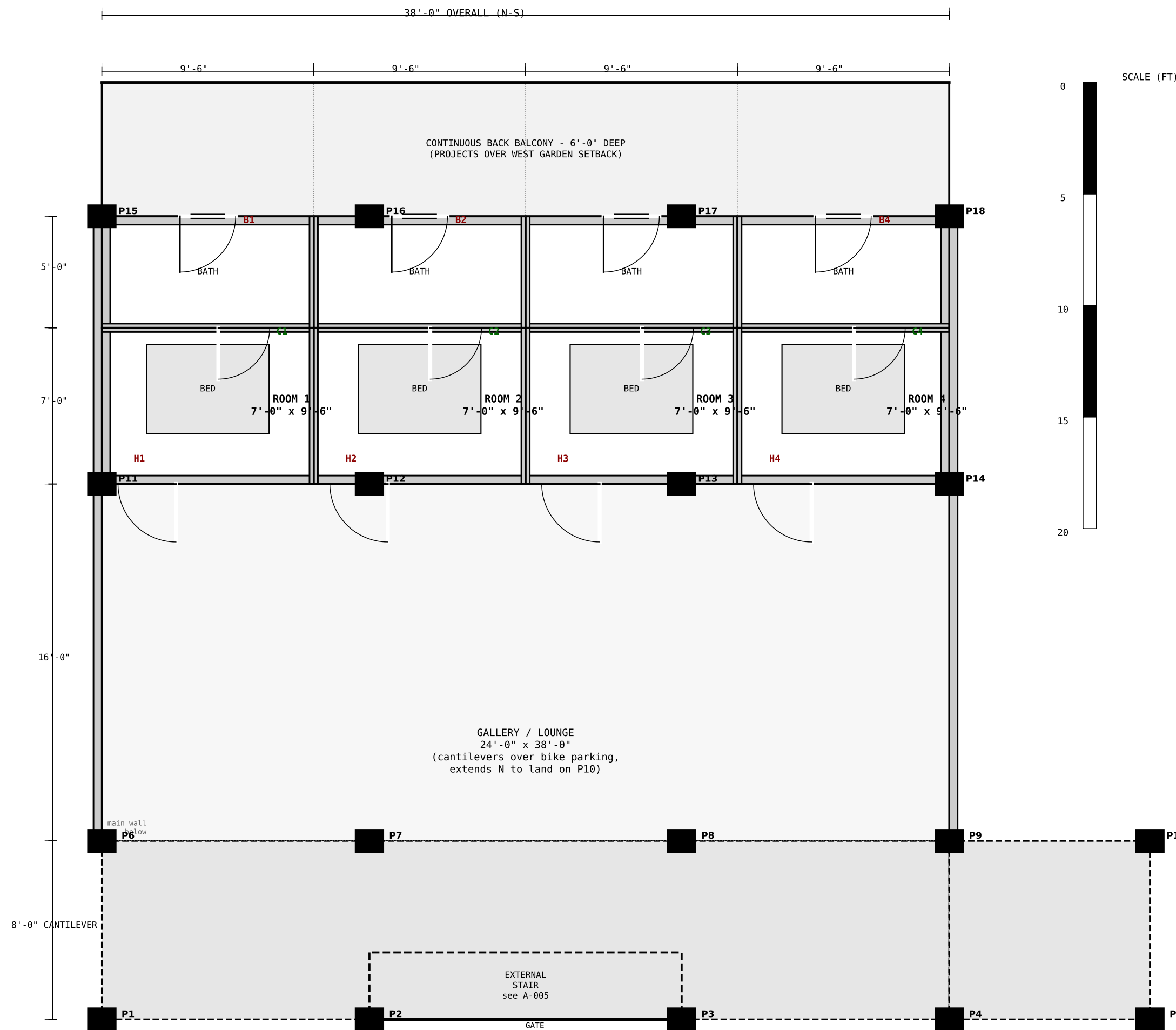
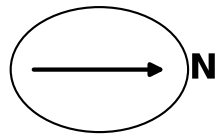
G+1+TERRACE RCC RESIDENTIAL (2 UNITS) – SCHEMATIC DESIGN / GFC-BASIS SET

SCALE: 1:75

Client: Owner Date: 20-09-2026 Not for construction without RCC/Architect stamp

DWG NO: A-002

SHEET
04



- NOTES:
- REVISION 8: Room resized to 12'-0" x 9'-6" total (bath 5'-0" + room 7'-0"), so the P11-P14 column line (x=12'-0") lands on the Room/Gallery wall, not inside the floor.
 - 4 independent rooms along the West/garden side, each with a 5'-0" x 6'-0" ensuite bath. Door labels: H = room entrance (from gallery), C = washroom entrance (from room), B = balcony door (from bath).
 - A continuous 6'-0" deep back balcony runs the full 38' West side, projecting over the West garden setback below; each room's bath opens onto it.
 - Gallery/lounge along the East side: 16' within the main footprint, cantilevering a further 8' to fully cover the bike-parking width below, and extending north to land squarely on the repositioned column P10.
 - REVISION 8: The common staircase (see Sheet A-005) is now 3'-0" wide (was 6'-0"), a single half-turn flight in the bike-parking zone between P2 and P3; it lands at the gallery's edge through a gated opening at x=36'.
 - Cantilever design (gallery beyond the main east wall, and the north extension to P10) must be checked and detailed by the structural engineer -- shown here schematically.
 - Wall thickness: external/load-bearing 9", internal partitions 4.5". Room doors 2'-6"x7'-0"; bath doors 2'-3"x7'-0"; balcony doors 2'-6"x7'-0".

PROPOSED G+1+TERRACE RESIDENTIAL BUILDING

FIRST FLOOR PLAN (4 ROOMS + GALLERY)

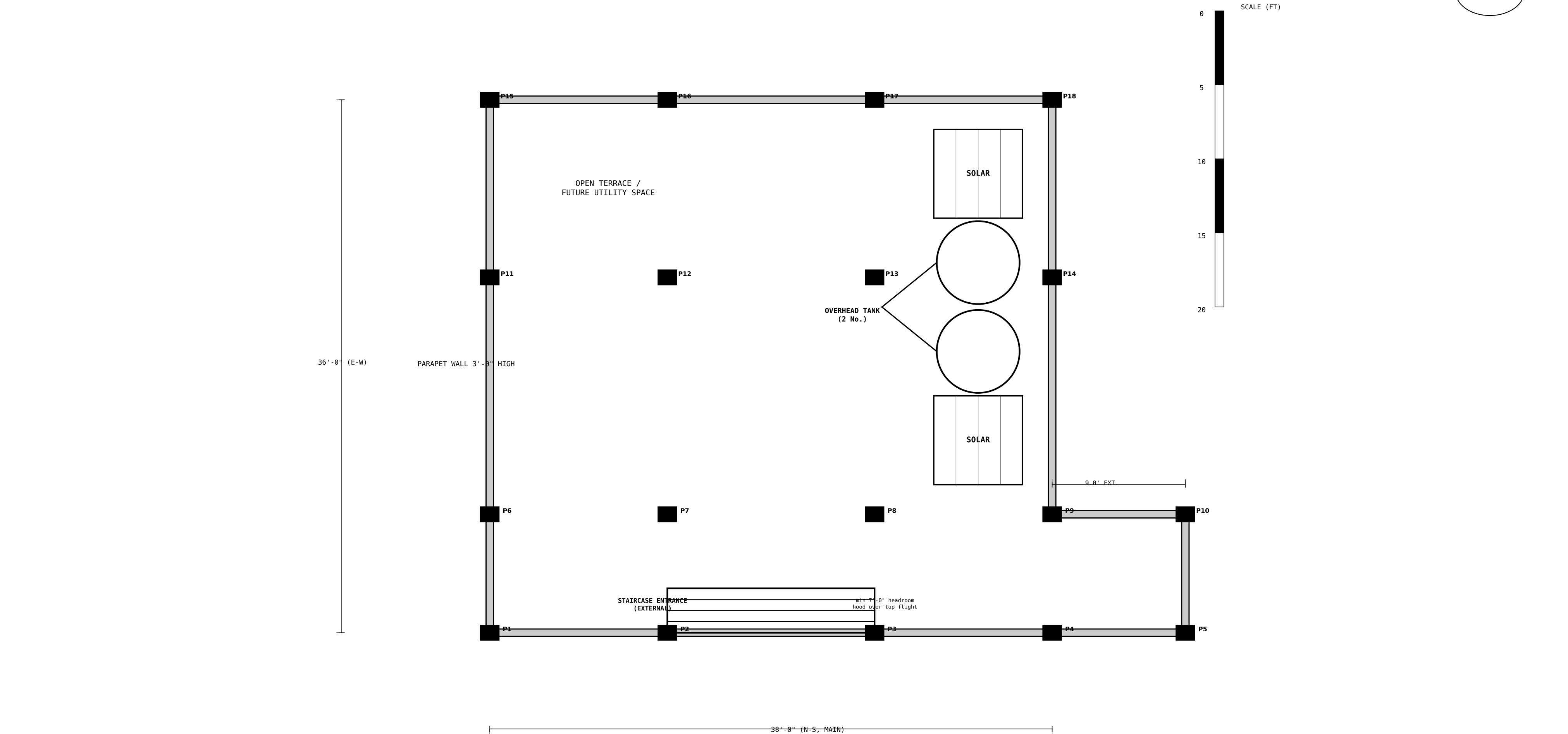
G+1+TERRACE RCC RESIDENTIAL (2 UNITS) – SCHEMATIC DESIGN / GFC-BASIS SET

SCALE: 1:75

Client: Owner Date: 20-09-2026 Not for construction without RCC/Architect stamp

DWG NO: A-003

SHEET
05



NOTES:

1. REVISION 8: Terrace slab matches the first floor below: main building + gallery cantilever over the bike-parking canopy, north extension covering only the cantilever width (P6/P9-P10 to canopy line), landing on P10.

2. The extra slab previously shown extending the north extension back to the P11-P14 line (i.e. between P10 and P14) has been removed per client instruction.

3. Parapet wall 3'-0" high all around the terrace edge, coping on top.

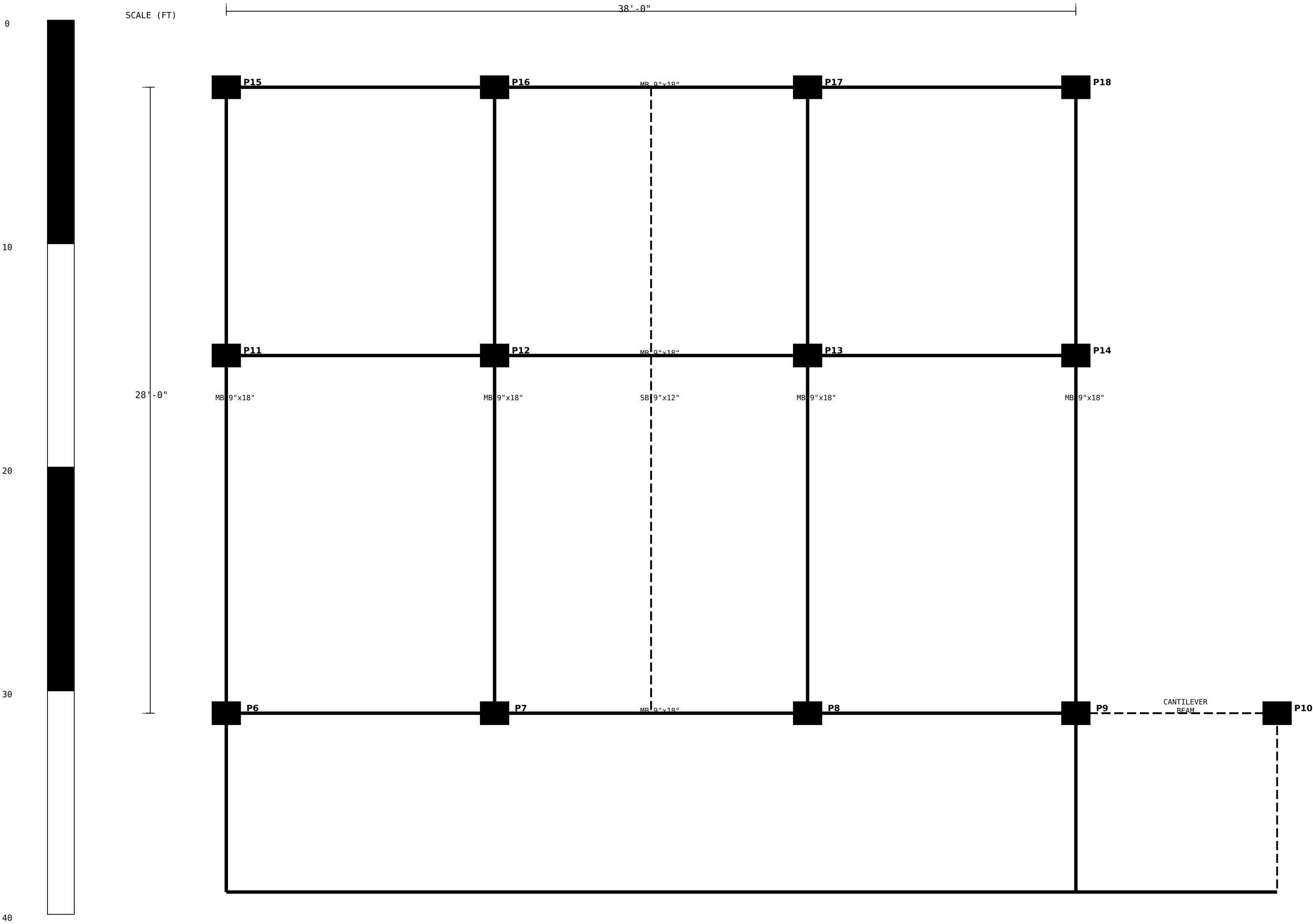
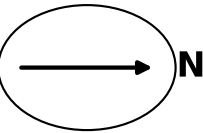
4. REVISION 8: Common staircase relocated OUTSIDE, in the bike-parking zone between columns P2 and P3 -- the original 14' bay (see Sheet A-005), now 3'-0" wide -- arriving at the terrace's East edge through a small headroom hood (min 7'-0" clear over the top of the stairs).

5. Two overhead RCC/sectional water tanks (each sized for a typical 1000-2000L), fed by a common riser from the common plumbing shaft below; verify combined capacity vs. daily demand for all 4 first-floor rooms + 2 ground-floor units.

6. Two Solar PV panel zones on raised MS mounting structures, tilted south-facing, flanking the tanks so neither shades the other; provide a hot-water line from either tank to a solar water heater if required.

7. Remaining terrace kept as open/future utility space; slope terrace surface (1:100 min) toward a proper rainwater drainage outlet with down-take pipe.

PROPOSED G+1+TERRACE RESIDENTIAL BUILDING	TERRACE FLOOR PLAN	
	G+1+TERRACE RCC RESIDENTIAL (2 UNITS) – SCHEMATIC DESIGN / GFC-BASIS SET SCALE: 1:75 Client: Owner Date: 20-09-2026 Not for construction without RCC/Architect stamp DWG NO: A-004	SHEET 06



NOTES:

1. Main beams 9" x 18" (width x overall depth) run along every column grid line, both directions, connecting all columns directly.
2. Secondary beams 9" x 12" support the mid-corridor partition (y=19, now a plain continuous wall), and the two cantilever beams P9-P10-P5 that carry the first-floor gallery's extension over the car-parking side.
3. REVISION 5: the common staircase is now external (bike-parking zone between P2 and P3) and is NOT part of this grid -- it has its own independent waist-slab, stringer beams and foundation; see Sheet A-005.
4. Beam depths are overall (including slab thickness); clear soffit depth to be confirmed by structural design.
5. This layout is for the Ground Floor roof (First Floor slab) level. The First Floor roof (Terrace slab) differs -- it must also support the gallery's cantilever over the bike-parking canopy (P1-P5 line) and the mummy -- verify independently, see Sheet A-005.
6. All beam sizes and reinforcement to be finalised by structural design/BBS; sizes shown are typical/preliminary for a G+1 RCC framed residence.

— MAIN BEAM 9"x18"
- - - SECONDARY BEAM 9"x12"

PROPOSED G+1+TERRACE RESIDENTIAL BUILDING

BEAM LAYOUT PLAN (OVER GROUND FLOOR)

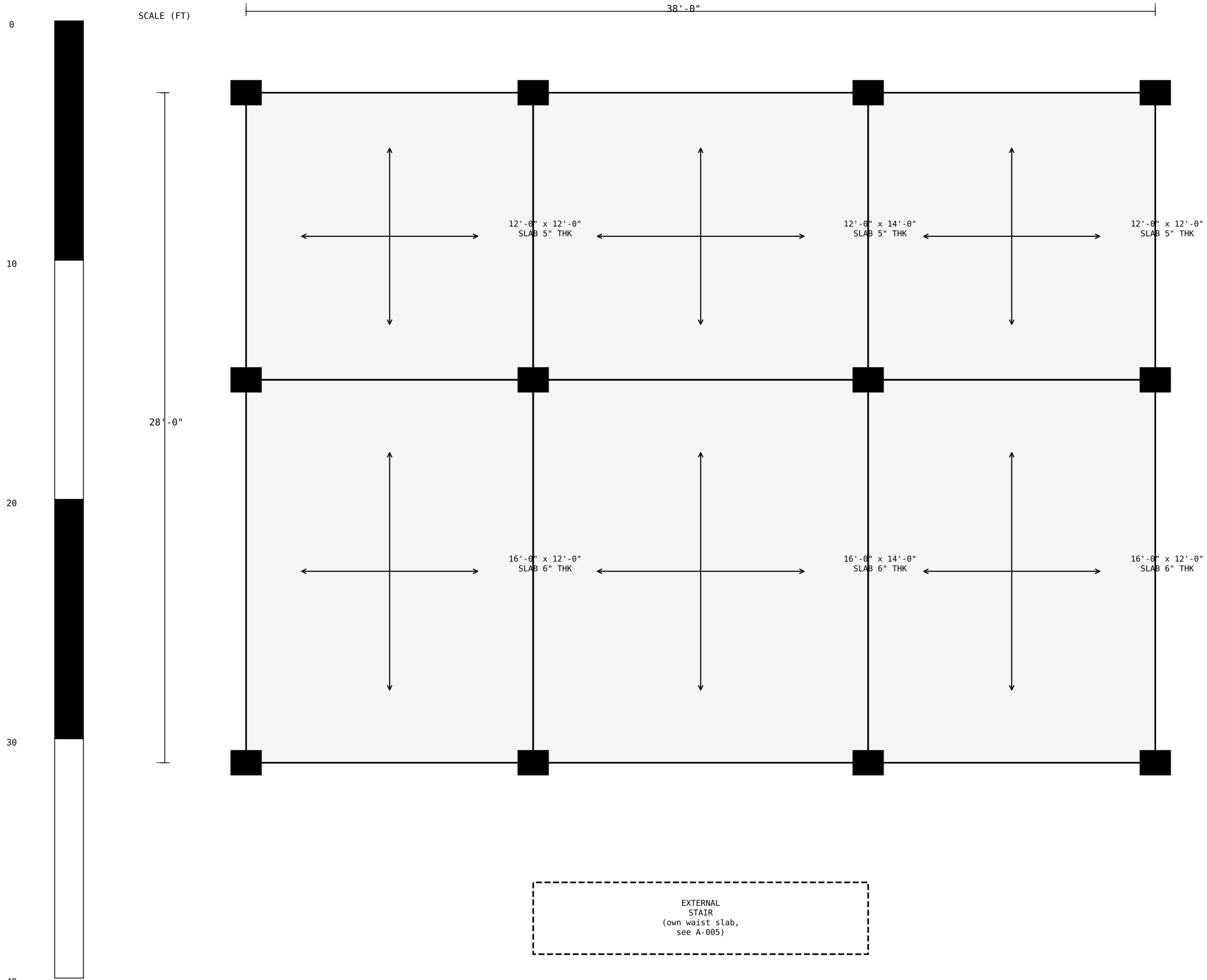
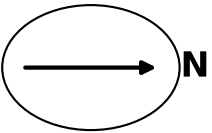
G+1+TERRACE RCC RESIDENTIAL (2 UNITS) – SCHEMATIC DESIGN / GFC-BASIS SET

SCALE: 1:100

SHEET

Client: Owner Date: 20-09-2026 Not for construction without RCC/Architect stamp DWG NO: S-003

07



- NOTES:
1. RCC slab thickness 5" (125mm) for panels under 180 sq.ft, 6" (150mm) for larger panels -- typical for a domestic two-way slab, confirm by design.
 2. Double-headed arrows indicate the two principal reinforcement (bottom steel) directions for each two-way slab panel; provide torsion steel at all discontinuous corners.
 3. Slab panels are bounded by the main/secondary beam grid shown on Sheet S-003. REVISION 5: the former internal stair opening is removed -- the corridor is now fully slabbed.
 4. The common staircase is now external (bike-parking zone, between P2 and P3) with its own independent waist slab -- see Sheet A-005; not part of this main-building slab.
 5. Provide extra top steel over all beam supports (negative moment zone), typically 0.3 x span length from the face of support, per structural design.
 6. M20 grade concrete, Fe500 grade reinforcement assumed typical; nominal cover 20mm (bottom) / 15mm (top) unless noted otherwise by the structural engineer.

PROPOSED G+1+TERRACE RESIDENTIAL BUILDING

SLAB LAYOUT & REINFORCEMENT DIRECTIONS

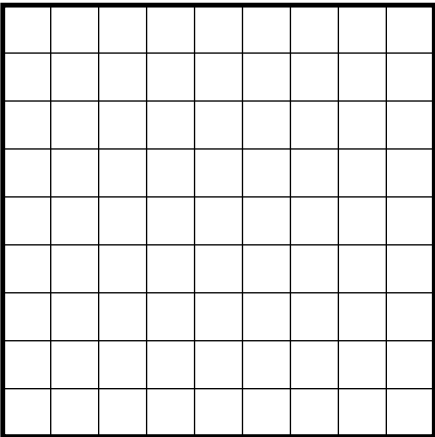
G+1+TERRACE RCC RESIDENTIAL (2 UNITS) – SCHEMATIC DESIGN / GFC-BASIS SET

SCALE: 1:100

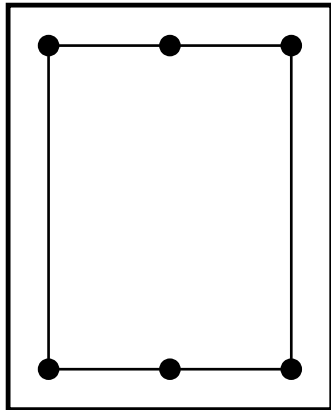
Client: Owner Date: 20-09-2026 Not for construction without RCC/Architect stamp DWG NO: S-004

SHEET
08

1. ISOLATED FOOTING - PLAN (BOTTOM MESH) 4. COLUMN CROSS-SECTION (12"x15")

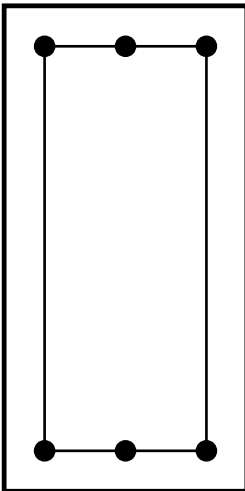


12mm dia @ 150mm c/c, BOTH WAYS, BOTTOM
6'x6'x1'-6" thk, bottom mesh only
Pedestal 2'-0"x2'-0": 6 No. 12mm main bars,
8mm ties @ 150mm c/c, full pedestal height



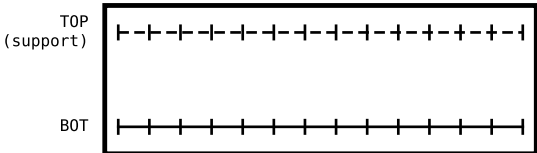
6 No. 16mm dia main bars (2 each face)
8mm dia ties @ 150mm c/c
(reduced to 100mm within 1x column-width of every joint)
Steel ratio = 1.04% of gross area

2. MAIN BEAM SECTION (9"x18")



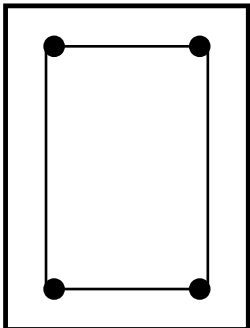
3 No. 16mm TOP + 3 No. 16mm BOTTOM
8mm dia stirrups @ 150mm c/c
(reduced to 100mm within 2x depth of every support)

3. SLAB SECTION (5"-6" THICK, TWO-WAY)



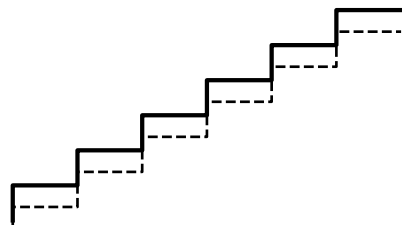
Main (bottom): 10mm dia @ 150mm c/c, shorter span
Distribution: 8mm dia @ 200mm c/c
Extra top steel over continuous supports, 0.3x span from face

5. SECONDARY BEAM SECTION (9"x12")



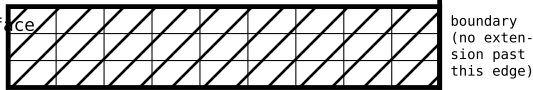
2 No. 12mm TOP + 2 No. 12mm BOTTOM
8mm dia stirrups @ 150mm c/c
Plinth/tie beams (9"x12"): same detail

6. STAIRCASE WAIST SLAB



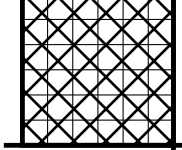
Main: 10mm dia @ 125mm c/c, along waist
Distribution: 8mm dia @ 200mm c/c, across width

7. HALF/EDGE FOOTING (7 No.)



P2, P3, P4, P6, P10, P11, P15
Column at the boundary-side edge; footing extends the full 3'-0" only into the plot (18 sq.ft, half the normal footing).
Bottom mesh: 12mm dia @ 150mm c/c, both ways
Tie/plinth beam acts as a STRAP BEAM to the next interior footing, resisting the eccentric moment -- engineer to confirm.

8. CORNER/QUARTER FOOTING (2 No.)



P1, P5
Trimmed on BOTH sides at the two true plot corners
(9 sq.ft, a quarter of the normal footing). Strap beams
both ways are essential -- structural engineer to confirm

- NOTES:
- Grades assumed: concrete M20, reinforcement Fe500 -- confirm with structural engineer.
 - All sizes shown are PRELIMINARY / thumb-rule detailing, consistent with typical practice for a low-rise (G+1) RCC-framed residence on rocky soil. Both the drawings and the quantities are for coordination and budgeting only.
 - Nominal cover: 50mm to footings, 40mm to columns, 25mm to beams, 15-20mm to slabs (IS 456 Table 16), unless the structural engineer specifies otherwise.
 - Lap lengths, development lengths, curtailment points and exact bar cutting/bending schedule (BBS) must be finalised by the structural engineer per IS 456 / SP 34, based on an actual analysis.
 - See Sheet S-004B for the corresponding preliminary steel quantity estimate.

PROPOSED G+1+TERRACE RESIDENTIAL BUILDING

TYPICAL REINFORCEMENT DETAILS

G+1+TERRACE RCC RESIDENTIAL (2 UNITS) – SCHEMATIC DESIGN / GFC-BASIS SET

SCALE: AS NOTED (schematic stamp)

Client: Owner Date: 20-09-2026 Not for construction without RCC/Architect stamp

DWG NO: S-004A

SHEET 08A

PRELIMINARY STEEL QUANTITY ESTIMATE

(thumb-rule quantities for budgeting/coordination -- NOT a substitute for a structural engineer's BBS)

MEMBER	QUANTITY / BASIS	BAR SIZE(S)	WEIGHT (kg)
Footings (18 No.: 9 full, 7 half, 2 corner)	402 bars, mixed lengths, 586m total	12mm dia	521
Pedestals (18 No.)	main 6x18 + ties 7x18	12mm main / 8mm ties	228
Columns (18 No., P1-P18)	main 6x18, ties 43x18, ht 6.4m	16mm main / 8mm ties	1,601
Main beams (9"x18"), 2 levels	289 ft/level x 2 levels = 176m run	16mm long. / 8mm stirrups	2,258
Secondary beams, 2 levels	90 ft run = 27m	12mm long. / 8mm stirrups	167
Plinth/tie beams, 1 level	226 ft run = 69m	12mm long. / 8mm stirrups	420
Slabs (FF + Terrace, incl. gallery)	2344 sqft = 218 m² @ 3.75kg/m²	10mm/8mm (see S-004A)	817
Staircase waist slab, 2 flights	63 sqft = 6 m² @ 5.0kg/m²	10mm/8mm (see S-004A)	29

GRAND TOTAL

6,040 kg

≈ 6.04 tonnes

Cross-check: 6,040 kg / ~2128 sqft built-up (GF+FF) ≈ 2.84 kg/sqft

Typical Indian practice for a G+1 RCC-framed residence: 3.5-4.5 kg/sqft including footing, column, beam & slab steel.

This estimate runs somewhat below that range because it uses minimum/typical bar counts and spacings appropriate to a light, low-rise structure on ROCKY soil (as assumed) -- a structural engineer's actual design, once wind/seismic zone and real loads are checked, commonly lands within or somewhat above the typical range. Budget with a margin.

NOTES:

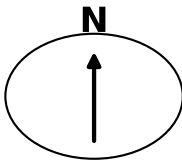
1. Steel grade Fe500, concrete grade M20 (assumed). Unit weights per IS 1786: w(kg/m) = dia²(mm)/162.

2. Quantities are derived directly from this drawing set's own grid/geometry (column count & height, beam/slab lengths & areas, footing/stair sizes) using the typical bar sizes and spacings shown on Sheet S-004A.

3. EXCLUDED: laps/development length beyond the single splice/joint per member, wastage (add 3-5%), chairs/spacers, and extra steel for the terrace mumty/parapet or the cantilever (once finalised, it will likely need more top steel than this allows).

4. This is a PRELIMINARY quantity for budgeting and material planning only. The actual bar bending schedule (BBS), covering every individual bar's exact cut length, bend points and count, must be produced by the structural engineer from a full structural analysis and design.

PRELIMINARY MATERIAL & COST ESTIMATE



(thumb-rule quantities for budgeting only -- rates are INDICATIVE, verify current local rates before use)

MATERIAL	QUANTITY	UNIT	RATE (Rs)	COST (Rs)
Steel reinforcement (Fe500 TMT)	6,040	kg	62	374,451
Cement (OPC/PPC, 50kg bags)	947	bags	400	378,859
Sand (fine aggregate)	77	cum	2,200	168,929
Aggregate / Khadi (20mm crushed stone)	80	cum	1,600	128,572
Bricks (common/clay, 19x9x9cm modular)	45,784	nos	8	366,269
Binding wire	60	kg	75	4,530
Material subtotal				1,421,610
LABOUR				
RCC labour (shuttering+steel-fixing+concreting)	2,344	sqft	220	515,680
General labour (brickwork+plastering)	152	days	1,100	166,755
Material transformation & slab-filling crew (4x Rs1100 + 4x Rs400)	24	days	6,000	143,495
Labour subtotal				825,931
GRAND TOTAL (material + labour, excl. finishes)				Rs 2,247,541

Material basis: RCC concrete 96 cum (M20, 1:1.5:3) | Brickwork 92 cum net (1:6 CM, 12% opening deduction) | Plaster 960 sqm net (12mm, 1:6)

Cement split: RCC 771 + brickwork 115 + plaster 60 bags. Sand split: RCC 40.2 + brickwork 24.1 + plaster 12.5 cum.

Labour basis (client-supplied rates, 2026-09): RCC labour = Rs 220/sqft x 2344 sqft RCC (slab) area cast.

General labour = Rs 1100/day x 152 labour-days, estimated from brickwork (92 days @ 1.0 cum/day) + plastering (60 days @ 16 sqm/day).

Material transformation/slab-filling crew = Rs 6000/day (4 x Rs1100 + 4 x Rs400) x 24 days, estimated from the 96 cum RCC concrete @ 4 cum/day placement rate for this crew.

Cross-check: Rs 2,247,541 / ~2128 sqft built-up (GF+FF) = Rs 1,056/sqft for STRUCTURAL MATERIAL + LABOUR ONLY.

Excludes shuttering/formwork hire, doors/windows, flooring, paint, plumbing & electrical fixtures, waterproofing, other trade labour, site supervision, contractor margin and GST.

NOTES:

1. MATERIAL rates are INDICATIVE ONLY (approx. national-average India, 2025) and WILL vary significantly by city, season, supplier and quantity ordered -- obtain current local quotations before budgeting; do not use for tendering.

2. LABOUR rates (Rs220/sqft RCC, Rs1100/day general, Rs1100+Rs400/day MT crew) are CLIENT-SUPPLIED (2026-09); all labour-DAY counts are this set's own estimate (CPWD-type norms), not client-supplied -- verify with your contractor.

3. POSSIBLE OVERLAP: the RCC labour line (covers shuttering+steel-fixing+CONCRETING) and the material-transformation/slab-filling crew (also covers concrete placement) may both price the same concreting work.

4. Confirm with your contractor whether these two are genuinely separate cost pools before totalling; if not, drop one of the two.

5. Steel quantity is carried directly from Sheet S-004B's preliminary BBS summary (same 18 footings/columns, beam/slab geometry). See S-004B for the member-by-member breakdown.

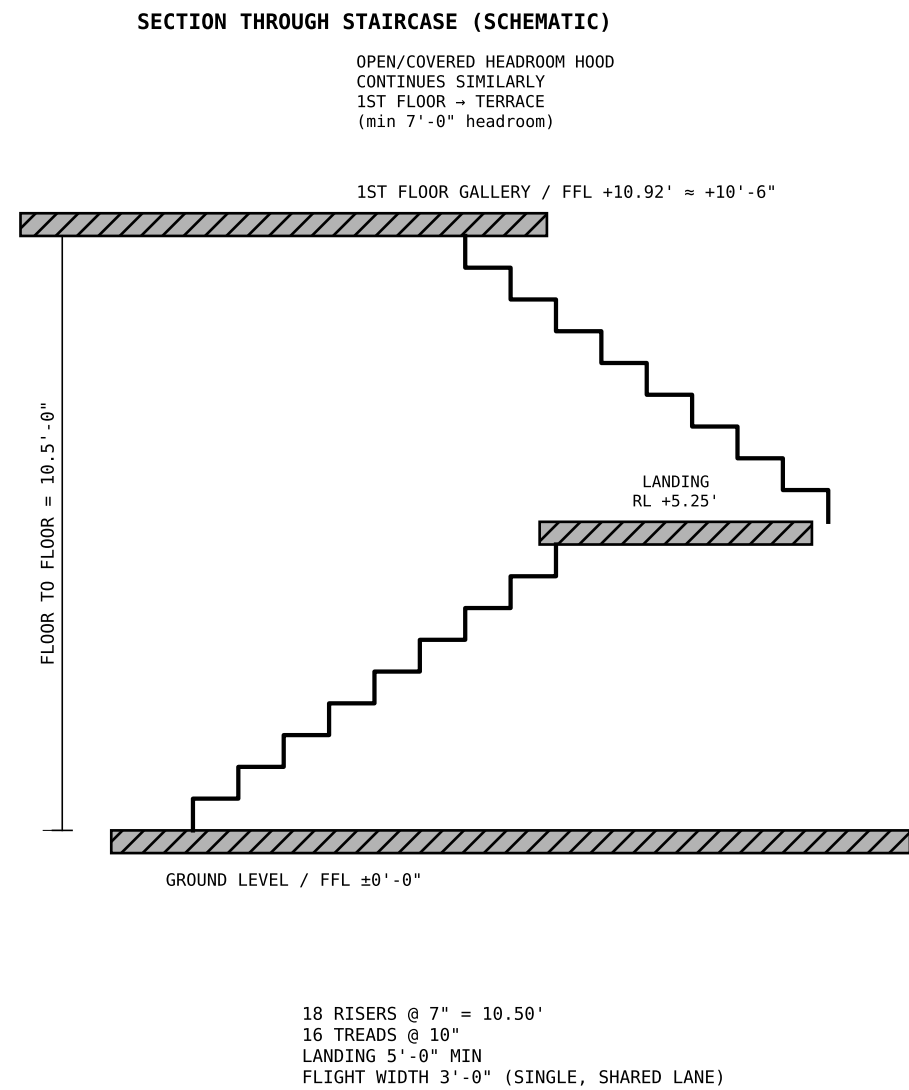
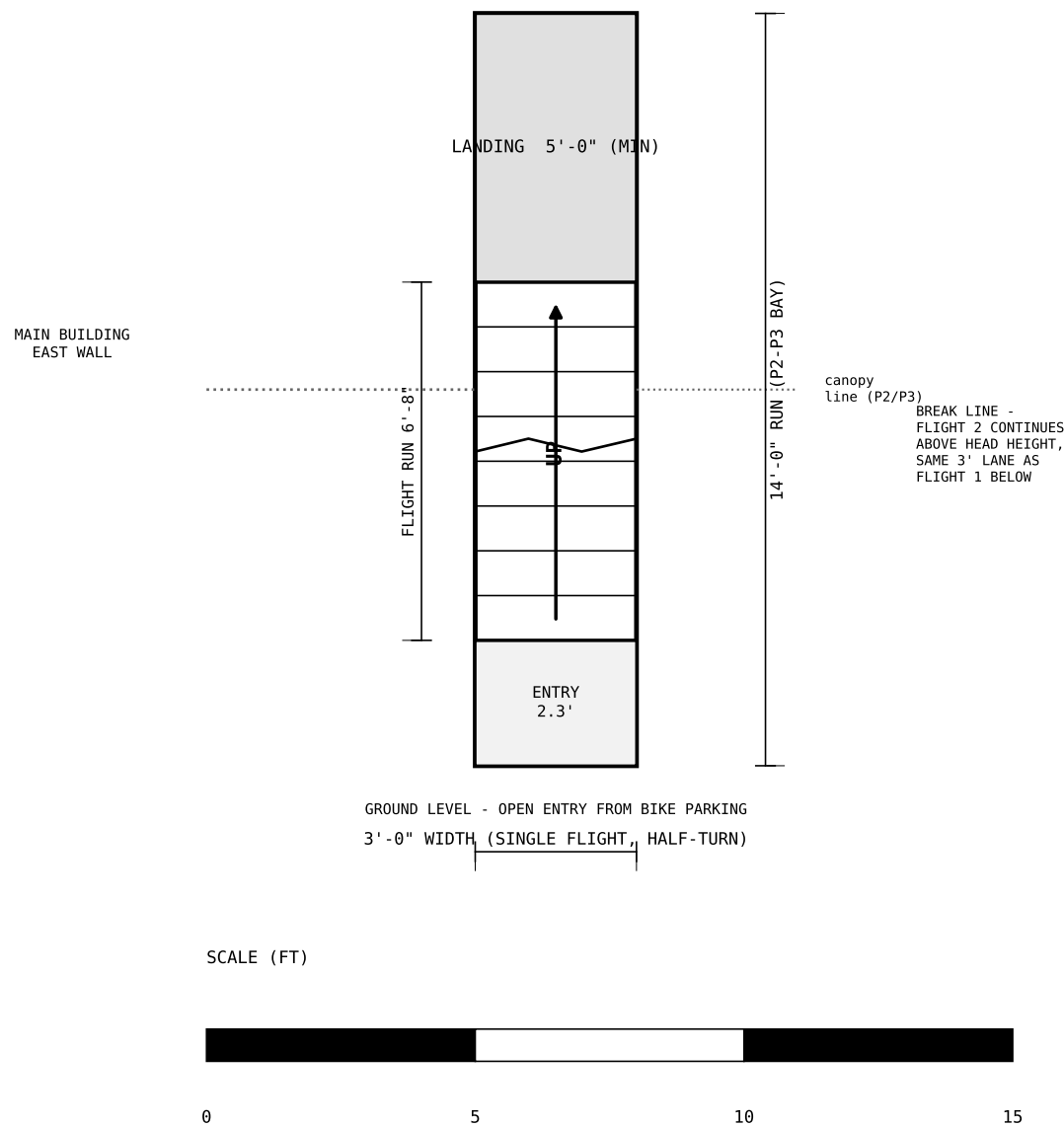
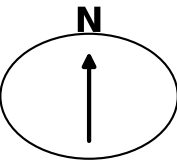
6. Concrete grade M20 (1:1.5:3 nominal mix) assumed for all RCC (footings, pedestals, columns, beams, slabs, waist slab); dry-volume factor 1.54 used for cement/sand/aggregate proportioning.

7. Brickwork assumed 1:6 cement:sand mortar, standard 19x9x9cm bricks (500/cum), covering external/common/internal walls (both floors) + parapet; a flat 12% deduction applied for openings -- a thumb-rule, not a door/window takeoff.

8. Plaster assumed 12mm thick, 1:6 cement:sand, both faces of every wall and the parapet, same 12% opening deduction.

9. EXCLUDED: shuttering/formwork, scaffolding, doors/windows, electrical, plumbing fixtures, sanitary ware, paint, flooring/tiling, water tank, skirting, waterproofing, site supervision, contractor margin/GST.

10. This is a PRELIMINARY budgeting estimate only. Final quantities and labour rates must be verified from the structural engineer's BBS (Sheet S-004B), your actual labour contractor, and a detailed architect's/contractor's BOQ before finalising a construction contract.



NOTES:

- REVISION 5: Common staircase is EXTERNAL (open, or covered by an extension of the bike-parking canopy), sited between columns P2 and P3, the original 14' bay -- exactly as marked up by the client.
- REVISION 8: Single-lane half-turn (scissor) stair, 3'-0" wide overall (was 6'-0", 2 flights side by side), connected by a 5'-0" half-space landing; Flight 2 climbs back over Flight 1's footprint, one level higher.
- Flight run + landing = 11'-8"; the remaining 2'-4" of the 14' bay is a small ground-level entry lobby, open to the bike parking.
- 18 risers @ 7" and 16 treads @ 10" for the 10'-6" floor-to-floor height (Ground to First Floor); the same scheme repeats First Floor to Terrace.
- Minimum headroom 7'-0" maintained throughout the flight and at the landing; provide a light roof/canopy extension (tying into the P1-P5 canopy) over the stair for weather protection.
- Provide a 3'-6" high handrail on the open side(s) of the flight and around the landing/lobby edge, plus a lockable gate at the ground-level entry for security.
- Being external, final position, guarding and the waist-slab/foundation design must be verified on site and by the structural engineer -- shown here schematically.

PROPOSED G+1+TERRACE RESIDENTIAL BUILDING

STAIRCASE PLAN & SECTION (COMMON)

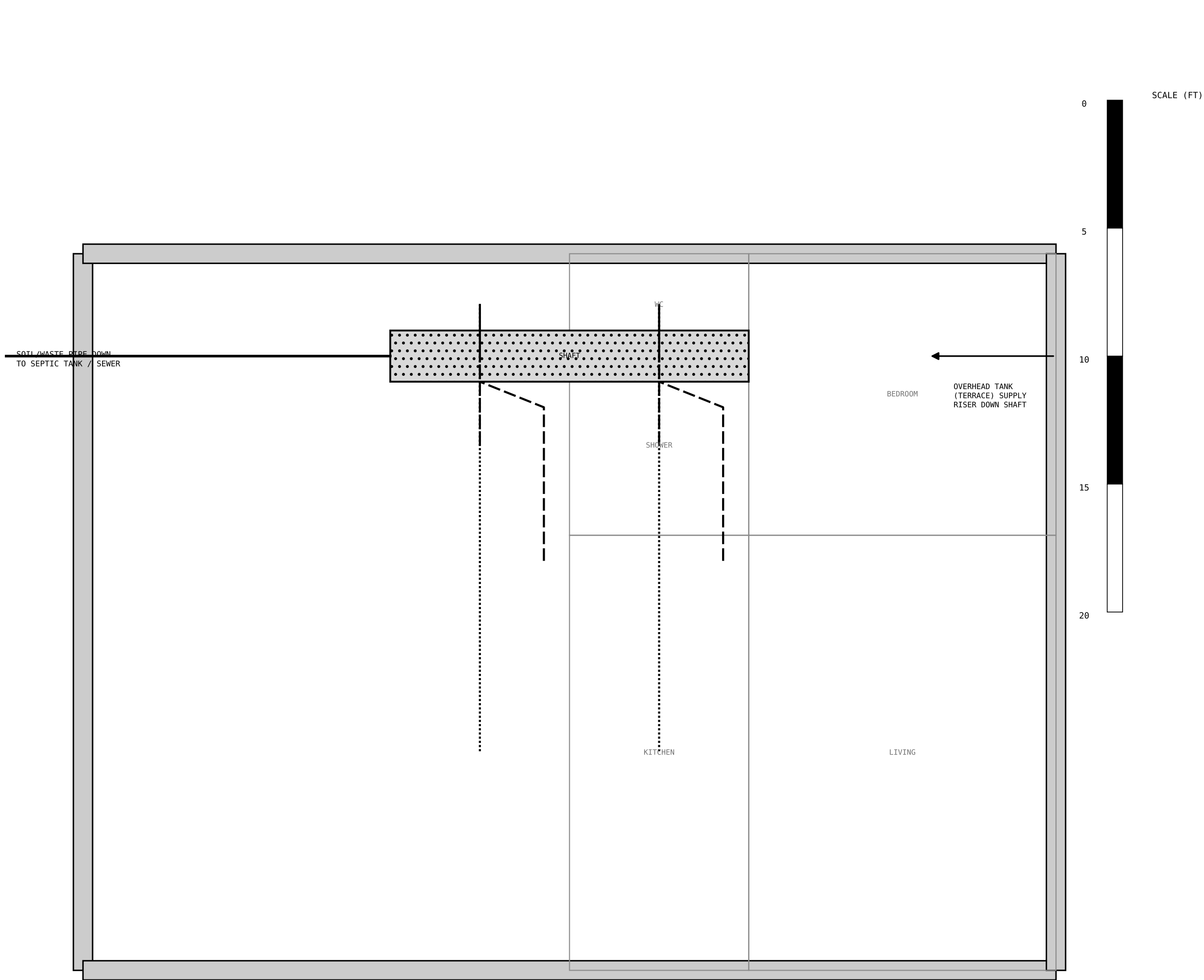
G+1+TERRACE RCC RESIDENTIAL (2 UNITS) – SCHEMATIC DESIGN / GFC-BASIS SET

SCALE: 1:50

Client: Owner Date: 20-09-2026 Not for construction without RCC/Architect stamp

DWG NO: A-005

SHEET
09



- NOTES:
1. Common plumbing shaft (2'x14') runs the full depth of the central corridor, aligning each unit's WC/Shower with the corresponding first-floor bath and the terrace overhead tank.
 2. Soil pipes (100mm dia. UPVC/CI) from each WC drop through the shaft to a common soil stack, terminating at a septic tank / soak pit or municipal sewer with vent pipe extended above roof level.
 3. Waste pipes (75mm dia.) from showers and kitchen sinks drop through the shaft to a waste stack, discharging to a gully trap and thence to the drainage system.
 4. Water supply: overhead tank on the terrace feeds a gravity riser down the shaft to all WC/Shower/Kitchen fixtures; provide isolation stop-cocks at each floor.
 5. Provide a separate cold-water rising main (with a ground-level sump + pump, if municipal pressure is insufficient) feeding the overhead tank.
 6. All pipe routes shown schematically; final sizing, gradients and inspection chamber locations to be finalised by the plumbing consultant.

- Soil pipe (WC)
- - Waste pipe (shower/kitchen)
- Water supply line

PROPOSED G+1+TERRACE RESIDENTIAL BUILDING

PLUMBING LAYOUT (WATER SUPPLY, SO

G+1+TERRACE RCC RESIDENTIAL (2 UNITS) – SCHEMATIC DESIGN / GFC-BASIS SET

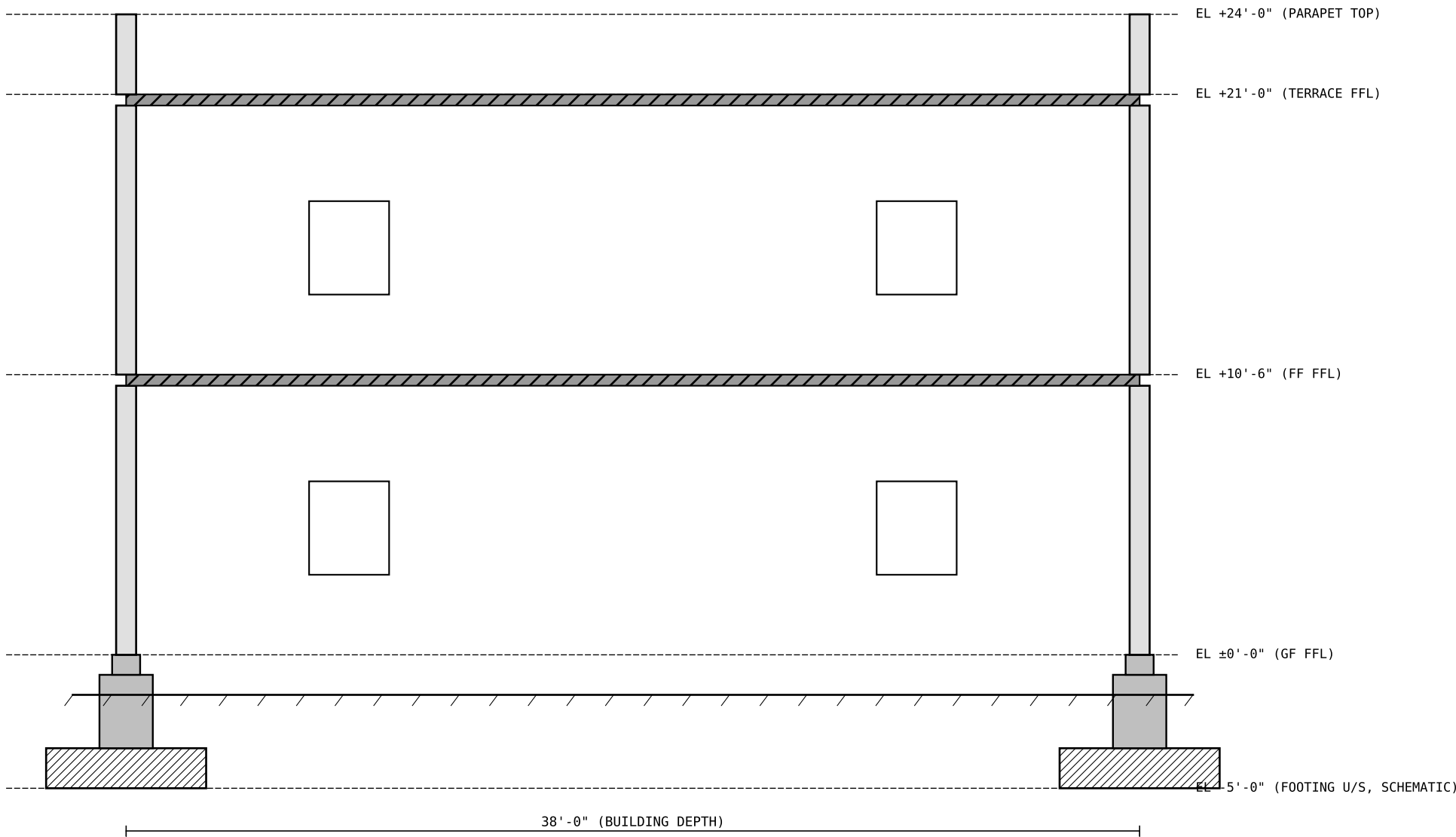
SCALE: 1:75

SHEET

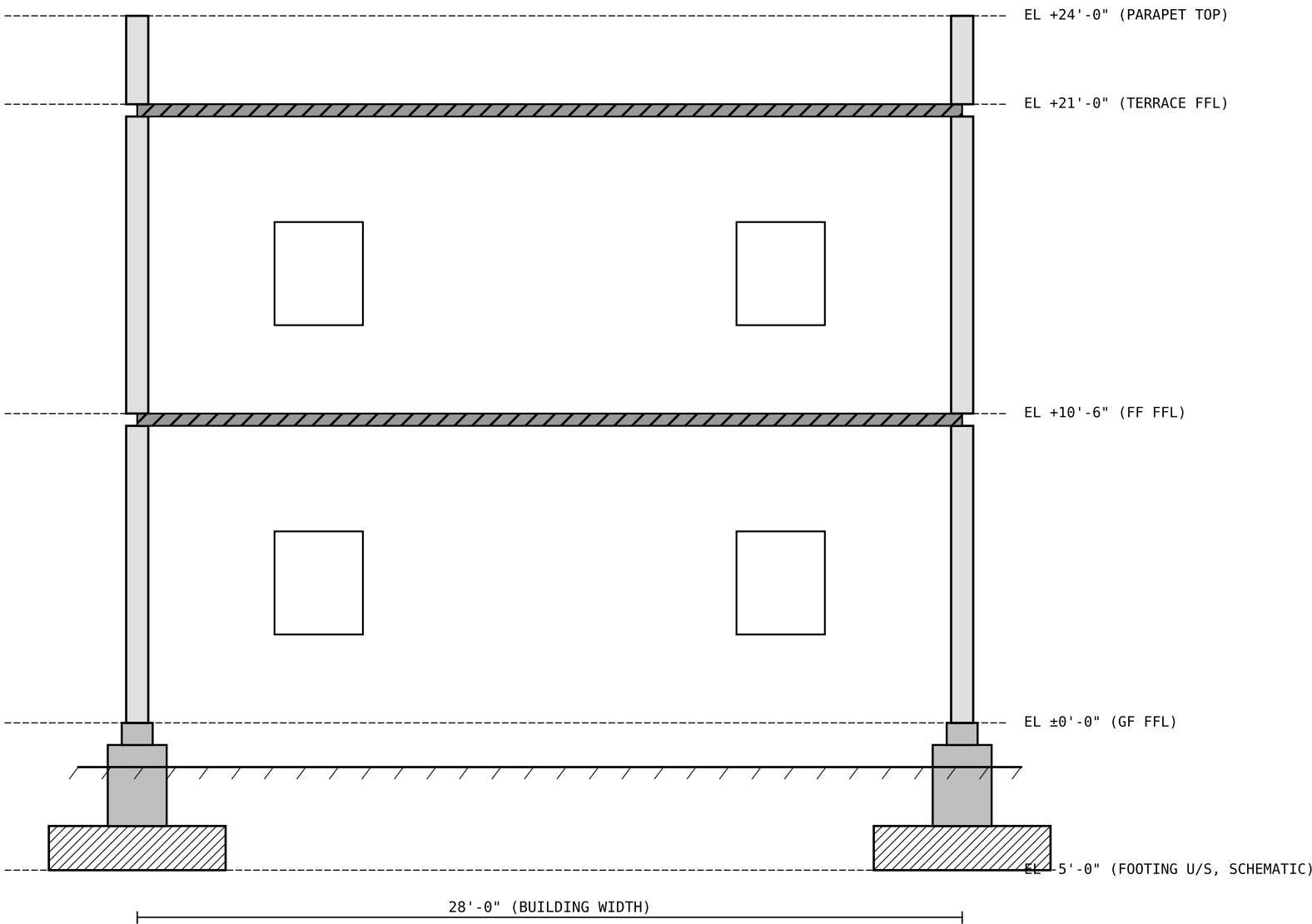
Client: Owner Date: 20-09-2026 Not for construction without RCC/Architect stamp DWG NO: P-001

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SECTION A-A (LOOKING EAST, THROUGH UNIT 2, N-S)



SECTION B-B (LOOKING NORTH, E-W)



- NOTES:
- Datum: Ground Floor finished floor level (FFL) taken as $\pm 0'-0"$. Founding/footing levels shown schematically; confirm actual founding depth on site (rocky strata) with the geotechnical/structural engineer.
 - Floor-to-floor height 10'-6" (Ground to First, and First to Terrace); slab thickness 5" typical, beam depths per Sheet S-003.
 - REVISION 5: the common staircase is now external (see Sheet A-005), in the bike-parking zone between P2 and P3 -- not shown in these two building sections, which cut through the main enclosed building only.
 - External walls 9" thick with plaster both sides; parapet wall 3'-0" high at terrace level.
 - Window head heights, sill levels and opening sizes shown schematically -- refer to Ground/First Floor plans for exact sizes and positions.

PROPOSED G+1+TERRACE RESIDENTIAL BUILDING

BUILDING SECTIONS A-A & B-B

G+1+TERRACE RCC RESIDENTIAL (2 UNITS) – SCHEMATIC DESIGN / GFC-BASIS SET

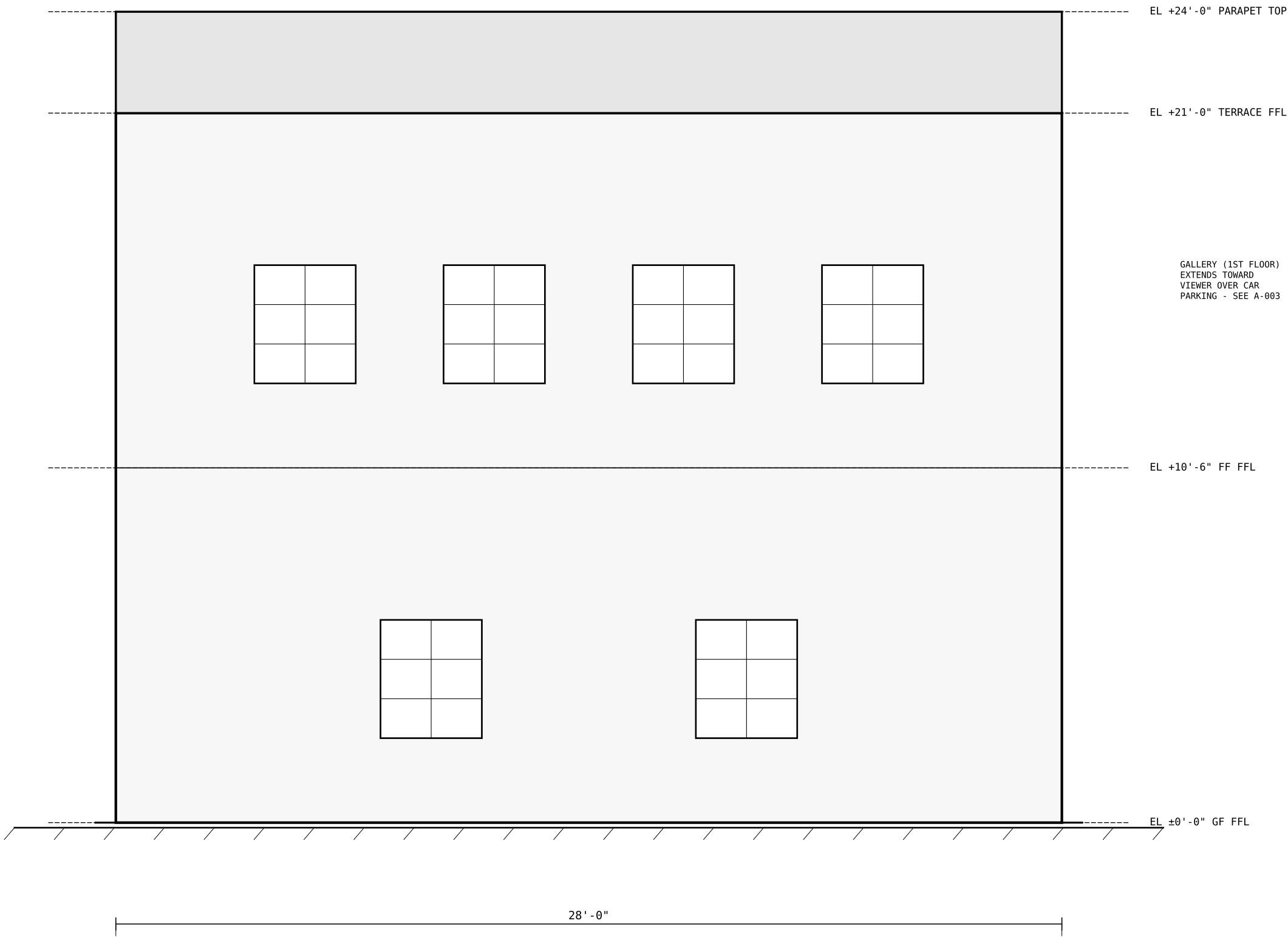
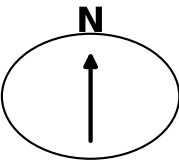
SCALE: 1:75

Client: Owner Date: 20-09-2026 Not for construction without RCC/Architect stamp

DWG NO: A-006

SHEET
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SCALE (FT)



NOTES:
1. North elevation faces the car-parking side; datum GF FFL taken as $\pm 0'-0"$ (matches Sheet A-006). Unit entries are on the East face -- see Sheet A-008.
2. REVISION 5: the common staircase and its headroom hood are external, on the East face (bike-parking side, between P2 and P3) -- not visible here, see Sheet A-008. The first-floor gallery extends toward the viewer over part of the car-parking setback to
3. The continuous back balcony is on the opposite (West/garden) face and is not visible in this elevation.
4. Window/door proportions indicative; exact sizes and positions per Ground/First Floor plans (Sheets A-002/A-003).
5. All RCC surfaces to receive weatherproof exterior paint/texture finish over sand-faced plaster; parapet coping with weathering slope and drip.
6. Final architectural treatment (cladding, sun-shades, colour scheme) to be developed at the design-development stage; this is a massing/schematic elevation.

PROPOSED G+1+TERRACE RESIDENTIAL BUILDING

FRONT ELEVATION (NORTH)

G+1+TERRACE RCC RESIDENTIAL (2 UNITS) – SCHEMATIC DESIGN / GFC-BASIS SET

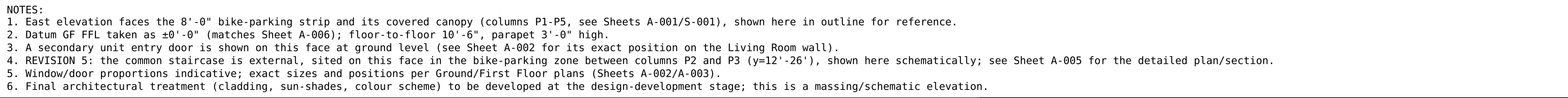
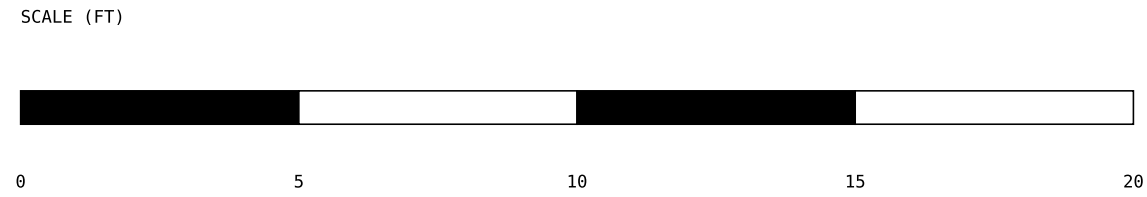
SCALE: 1:75

Client: Owner Date: 20-09-2026 Not for construction without RCC/Architect stamp

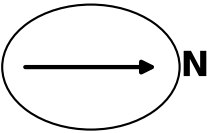
DWG NO: A-007

SHEET

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PROPOSED G+1+TERRACE RESIDENTIAL BUILDING		SIDE ELEVATION (EAST)
G+1+TERRACE RCC RESIDENTIAL (2 UNITS) – SCHEMATIC DESIGN / GFC-BASIS SET SCALE: 1:75 Client: Owner Date: 20-09-2026 Not for construction without RCC/Architect stamp		SHEET 14



DOOR SCHEDULE

D1	Unit Entry (E)	3'-0" x 7'-0"	Qty:1
D2	Unit 2 -> Ext. Stair	3'-0" x 7'-0"	Qty:1
D3	Internal	2'-6" x 7'-0"	Qty:6
D4	WC/Shower	2'-3" x 7'-0"	Qty:8
D5	Bedroom -> WC (back)	3'-0" x 7'-0"	Qty:2

WINDOW SCHEDULE

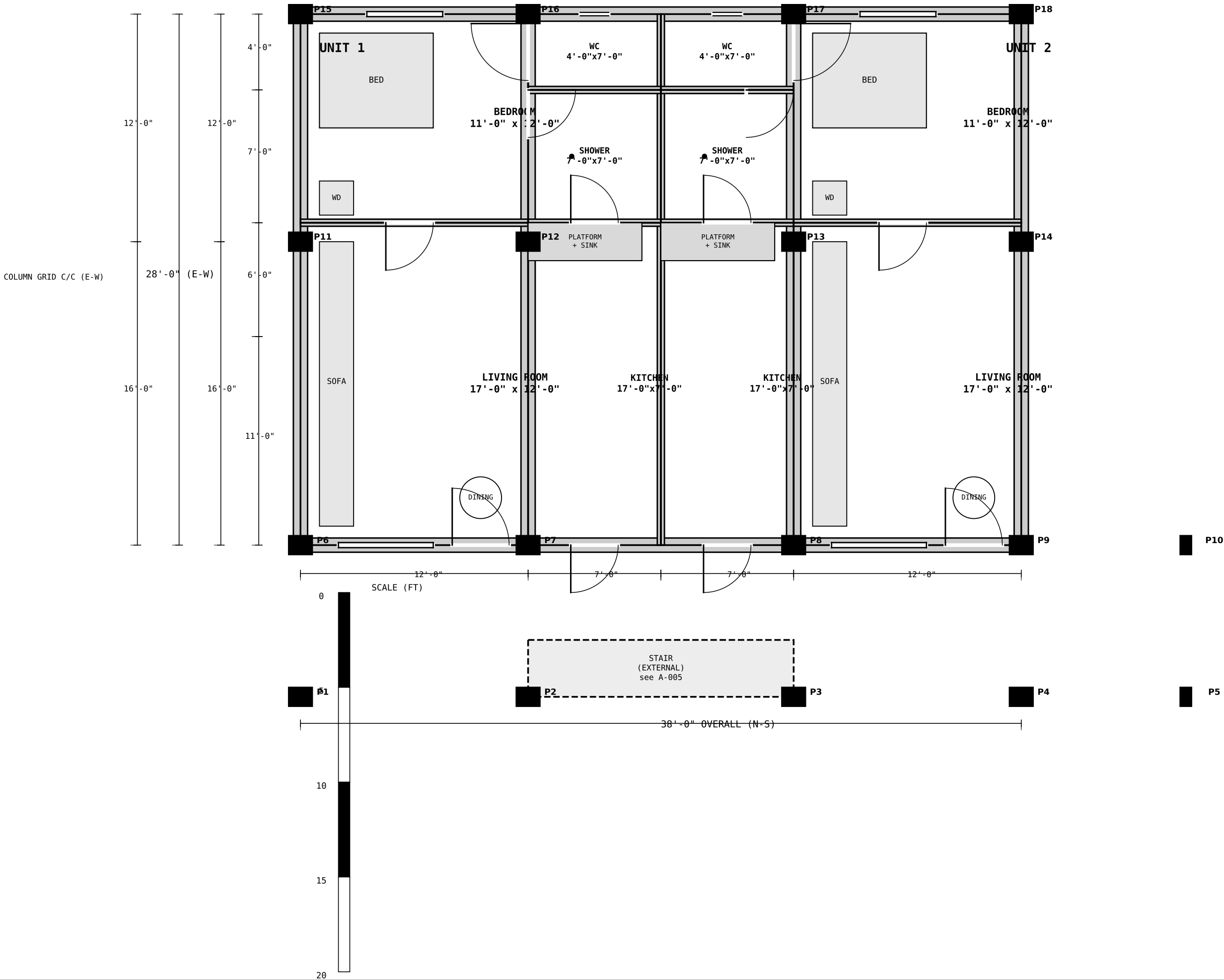
W1	Living Rm	5'-0" x 4'-6"	Qty:2
W2	Bedroom	4'-0" x 4'-0"	Qty:2
V1	WC/Shower Ventilator	1'-6" x 1'-6"	Qty:4

COLUMN & FOOTING SCHEDULE

Columns: P1-P18, 18 No., 12"x15" RCC
P10: corner column, carries 1st-floor
gallery extension over car parking
Footings: 18 No. isolated, 1'-6" thk. 9 full
6'x6'; 7 HALF (3'x6'/6'x3') + 2 CORNER
(3'x3') at boundary -- see Sht S-004A
Pedestals: 2'x2' RCC up to plinth
Main beams: 9"x18" | Secondary: 9"x12"
Slab: 5"-6" thk RCC, M20/Fe500 (typ.)

WALL LEGEND

- External wall - 9"
- Internal partition - 4.5"
- Common / party wall - 9"



NOTES:

- This master sheet consolidates the Ground Floor Plan (Sheet A-002) with complete centre-to-centre and clear dimensions, plus door/window/column schedules, in a single AutoCAD-style dimensioned blueprint.
- All dimensions in feet-inches unless noted; read in conjunction with Sheets A-001 through A-008 and S-001 through S-004 for full coordination.
- Refer to Sheet A-000 for the design-basis reconciliation notes -- several brief dimensions were adjusted marginally to produce a consistent, buildable set.

PROPOSED G+1+TERRACE RESIDENTIAL BUILDING

DIMENSIONED GROUND-FLOOR BLUEPRINT

G+1+TERRACE RCC RESIDENTIAL (2 UNITS) – SCHEMATIC DESIGN / GFC-BASIS SET

SCALE: 1:75 (AutoCAD-style) SHEET

Client: Owner Date: 20-09-2026 Not for construction without RCC/Architect stamp DWG NO: A-009

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